WordCount.java

```
import java.io.IOException; import
java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
 import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
  public static class TokenizerMapper extends
      Mapper<Object, Text, Text, IntWritable>{

    private final static IntWritable one = new IntWritable(1); private
    Text word = new Text();

    public void map(Object key, Text value, Context context
                    ) throws IOException, InterruptedException {
      StringTokenizer itr = new StringTokenizer(value.toString());
      while (itr.hasMoreTokens()) { word.set(itr.nextToken());
        context.write(word, one);
      }
    }
  }
public static class IntSumReducer extends
      Reducer<Text,IntWritable,Text,IntWritable> { private
    IntWritable result = new IntWritable();

    public void reduce(Text key, Iterable<IntWritable> values,
                       Context context
                       ) throws IOException, InterruptedException { int sum
      = 0;
      for (IntWritable val : values) {
        sum += val.get();
      }
      result.set(sum);
      context.write(key, result);
    }
  }

  public static void main(String[] args) throws Exception
    { Configuration conf = new Configuration(); Job job =
    Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
```

```
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new
    Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0: 1);
  }
}
```

Input:-

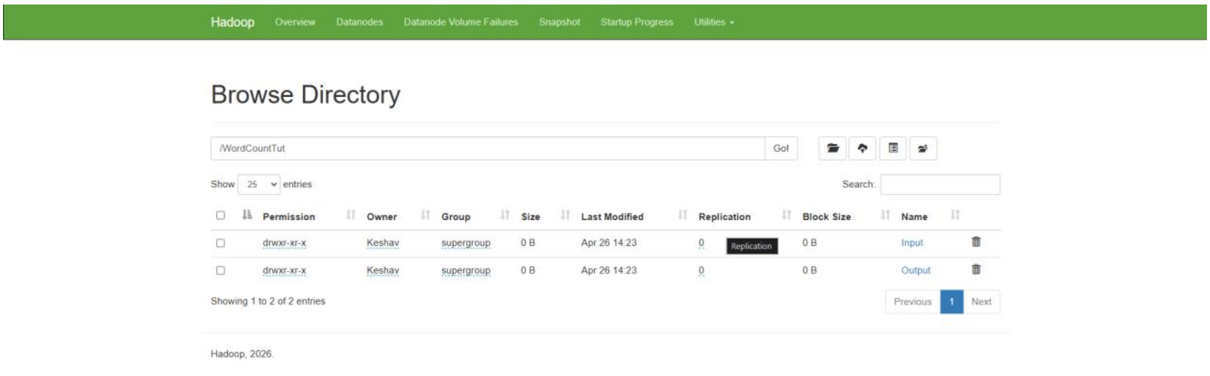
```
Generative AI and Large Language Models are transforming the landscape of software
engineering. By utilizing advanced algorithms and vast amounts of data, these models can
generate code, debug complex backend systems, and assist in architecting high-performance
applications. Retrieval-Augmented Generation, or RAG, enhances these models by connecting them
to external knowledge bases. This means an AI agent can read through thousands of PDFs,
extract embeddings, and query vector databases to provide highly accurate and contextual
answers.
```

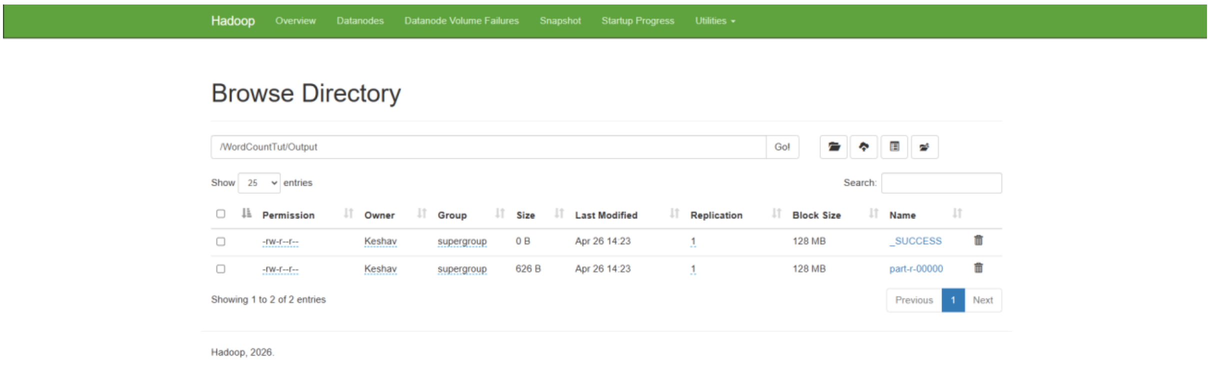
Algo steps :-

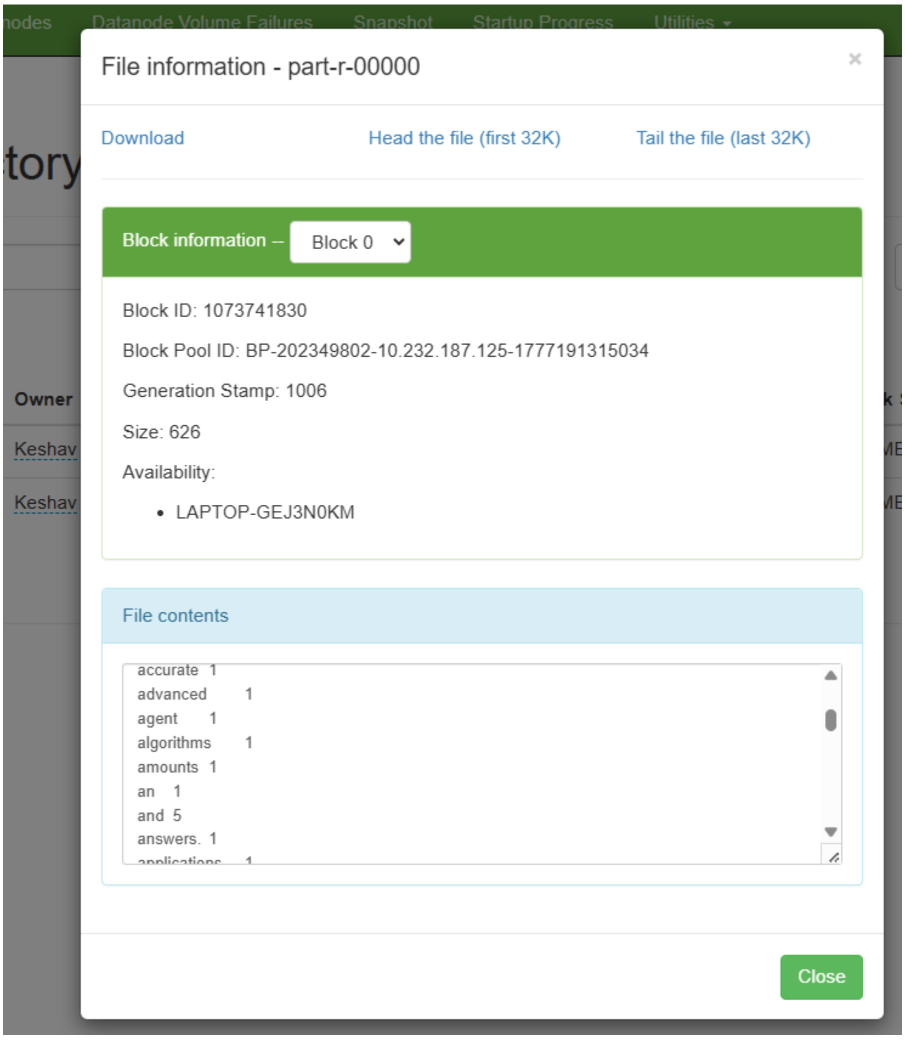
```
C:\WordCountExp>hadoop jar wcjar.jar WordCount /WordCountTut/Input /WordCountTut/Output
2026-04-26 14:23:51,593 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and exec
2026-04-26 14:23:51,662 WARN mapreduce.JobResourceUploader: No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
2026-04-26 14:23:51,764 INFO input.FileInputFormat: Total input files to process : 1
2026-04-26 14:23:52,048 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-26 14:23:52,333 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1997963917_0001
2026-04-26 14:23:52,335 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-26 14:23:52,638 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-26 14:23:52,641 INFO mapreduce.Job: Running job: job_local1997963917_0001
2026-04-26 14:23:52,645 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-26 14:23:52,656 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-04-26 14:23:52,658 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-26 14:23:52,663 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore c
2026-04-26 14:23:52,663 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-04-26 14:23:52,740 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-26 14:23:52,741 INFO mapred.LocalJobRunner: Starting task: attempt_local1997963917_0001_m_000000_0
2026-04-26 14:23:52,772 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-04-26 14:23:52,772 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-26 14:23:52,773 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore c
2026-04-26 14:23:52,780 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.
2026-04-26 14:23:52,811 INFO mapred.Task:  Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBasedProcessTree@5077cc9d
```

```
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=379584512
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=542
        File Output Format Counters
                Bytes Written=626
```

```
C:\WordCountExp>jar -cvf wcjar.jar WordCount*.class
added manifest
adding: WordCount$IntSumReducer.class(in = 1793) (out= 774)(deflated 56%)
adding: WordCount$TokenizerMapper.class(in = 1790) (out= 791)(deflated 55%)
adding: WordCount.class(in = 1511) (out= 833)(deflated 44%)
```







File information - part-r-00000
Download
Head the file (first 32K)
Tail the file (last 32K)
Block information -- Block 0
Block ID: 1073741830
Block Pool ID: BP-202349802-10.232.187.125-1777191315034
Generation Stamp: 1006
Size: 626
Availability:
LAPTOP-GEJ3N0KM
File contents
accurate 1
advanced 1
agent 1
algorithms 1
amounts 1
an 1
and 5
answers. 1
Close

Output:-

```
C:\WordCountExp>hdfs dfs -cat /WordCountTut/Output/part-r-00000
AI      2
By      1
Generation,     1
Generative      1
Language        1
Large   1
Models  1
PDFs,   1
RAG,    1
Retrieval-Augmented     1
This    1
accurate        1
advanced        1
agent   1
algorithms      1
amounts 1
an      1
and     5
answers.        1
applications.   1
architecting    1
are     1
assist  1
backend 1
bases.  1
by      1
can     2
code,   1
complex 1
connecting      1
contextual      1
data,   1
databases       1
debug   1
embeddings,     1
engineering.    1
enhances        1
external        1
extract 1
generate        1
high-performance        1


knowledge       1
landscape       1
means   1
models  2
of      3
or      1
provide 1
query   1
read    1
software        1
systems,        1
the     1
them    1
these   2
thousands       1
through 1
to      2
transforming    1
utilizing       1
vast    1
vector  1
```

1> **LogFileMapper.java** (Use for mapping the IP addresses from input csv file)

```
package LogFileCountry;
importjava.io.IOException;

import org.apache.hadoop.io.IntWritable;
import rg.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapred.*;

public class LogFileMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text,
IntWritable> { private final static IntWritable one = new
        IntWritable(1);

        public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output,
Reporter reporter) throws IOException {

                String  valueString  =  value.toString();  String[]
                SingleIpData        =        valueString.split("-");
                output.collect(new        Text(SingleIpData[0]),
                one);
        }
}
```

2>**LogFileReduce.java** (Use for reducing data received from mapper process to final output)

```
package LogFileCountry; import java.io.IOException; import java.util.*;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapred.*;

public class LogFileReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text,
IntWritable> {
        public void reduce(Text t_key, Iterator<IntWritable> values, OutputCollector<Text,IntWritable>
output, Reporter reporter) throws IOException {
                Text key = t_key;
                int frequencyForIp = 0;
                while (values.hasNext()) {
                        // replace type of value with the actual type of our value
                        IntWritable    value    =    (IntWritable)    values.next();
                        frequencyForIp += value.get();
                }
                output.collect(key, new IntWritable(frequencyForIp));
        }
}
```

3>**LogFileCountryDriver.java** (The driver code to run map-reduce on hdfs)

```
package LogFileCountry;

import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
importorg.apache.hadoop.mapred.*;

public class LogFileCountryDriver {
	public static void main(String[] args) {
		JobClient my_client = new JobClient();
		// Create a configuration object for the job
		JobConf job_conf = new JobConf(LogFileCountryDriver.class);

		// Set a name of the Job
		job_conf.setJobName("LogFileIP"); // Specify
		data type of output key and value
		job_conf.setOutputKeyClass(Text.class);
		job_conf.setOutputValueClass(IntWritable.class)
		; // Specify names of Mapper and Reducer Class
		job_conf.setMapperClass(LogFileCountry.LogFileMapper.class);
		job_conf.setReducerClass(LogFileCountry.LogFileReducer.class);

		// Specify formats of the data type of Input and output
		job_conf.setInputFormat(TextInputFormat.class);
		job_conf.setOutputFormat(TextOutputFormat.class);

		// Set input and output directories using command line arguments,
		//arg[0] = name of input directory on HDFS, and arg[1] = name of output directory to be
created to store the output file.

		FileInputFormat.setInputPaths(job_conf, new Path(args[0]));
		FileOutputFormat.setOutputPath(job_conf, new Path(args[1]));

		my_client.setConf(job_conf);
		try { // Run the job
		JobClient.runJob(job_conf);
		} catch (Exception e) {
			e.printStackTrace();
		}
	}
}
```

4> **log_file.txt** (Input file sample)

```
0.223.157.186  -  -  [15/Jul/2009:20:50:32  -0700]  "GET  /assets/js/the-associates.js  HTTP/1.1"  304
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/home-logo.png HTTP/1.1" 304 -
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/primary-news-2.jpg HTTP/1.1" 304
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/primary-news-1.jpg HTTP1.1" 304
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/home-media-block-placeholder.jpg HTTP/1.1" 304
```

-

10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/secondary-news-4.jpg HTTP/1.1" 304 10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/loading.gif HTTP/1.1" 304 10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/search-button.gif HTTP/1.1" 304 –

Input:-

```
log_file.txt
File  Edit  View
10.223.157.186 - - [15/Jul/2009:20:50:32 -0700] "GET /assets/js/the-associates.js HTTP/1.1"
304 -
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/home-logo.png HTTP/1.1" 304 -
10.1.232.31 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/primary-news-2.jpg
HTTP/1.1" 304 -
10.103.190.81 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/primary-news-1.jpg
HTTP1.1" 304 -
10.223.157.186 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/home-media-block-
placeholder.jpg HTTP/1.1" 304 -
10.1.232.31 - - [15/Jul/2009:20:50:33 -0700] "GET /assets/img/dummy/secondary-news-4.jpg
HTTP/1.1" 304 -
```

Algo:-

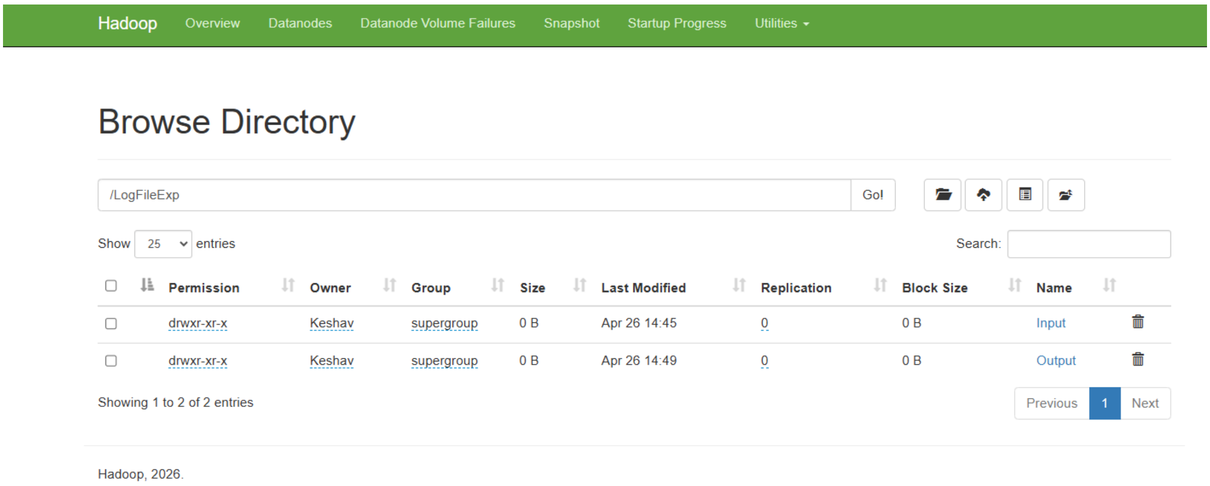
```
C:\LogFileExp>for /f "delims=" %i in ('hadoop classpath') do set HADOOP_CLASSPATH=%i

C:\LogFileExp>set HADOOP_CLASSPATH=C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common;C:\hadoop\share\hadoo
p\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\had
hare\hadoop\common\lib\*;C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop
n\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\mapreduce\*;
```

```
C:\LogFileExp>jar -cvf analyzelogs.jar LogFileCountry\*.class
added manifest
adding: LogFileCountry/LogFileCountryDriver.class(in = 1668) (out= 826)(deflated 50%)
adding: LogFileCountry/LogFileMapper.class(in = 1755) (out= 683)(deflated 61%)
adding: LogFileCountry/LogFileReducer.class(in = 1619) (out= 672)(deflated 58%)
```

```
C:\LogFileExp>hadoop jar analyzelogs.jar LogFileCountry.LogFileCountryDriver /LogFileExp/Input /LogFileExp/Output
2026-04-26 14:49:51,563 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-26 14:49:51,784 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interfa
2026-04-26 14:49:51,869 WARN mapreduce.JobResourceUploader: No job jar file set.  User classes may not be found. See Job or Job#setJar(S
2026-04-26 14:49:51,976 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-26 14:49:52,253 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-26 14:49:52,606 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local143983863_0001
2026-04-26 14:49:52,614 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-26 14:49:52,999 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-26 14:49:53,000 INFO mapred.LocalJobRunner: OutputCommitter set in config null
```

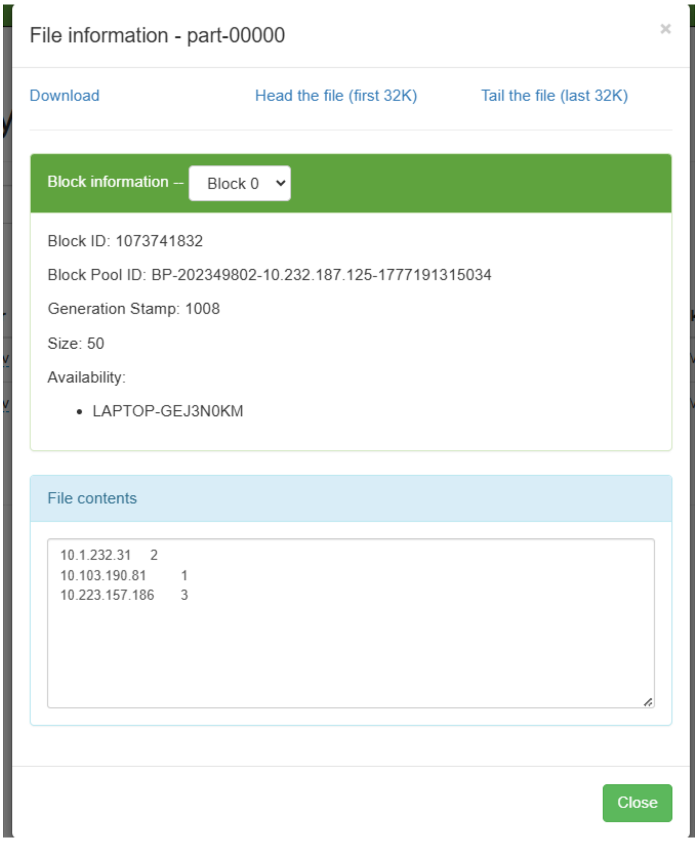
```
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=621
        File Output Format Counters
                Bytes Written=50
```

```
C:\LogFileExp>dir
 Volume in drive C is Windows
 Volume Serial Number is EA37-06AE

 Directory of C:\LogFileExp

26-04-2026  14:49    <DIR>          .
26-04-2026  14:49             3,017 analyzelogs.jar
26-04-2026  14:48    <DIR>          LogFileCountry
26-04-2026  14:45               621 log_file.txt
               2 File(s)          3,638 bytes
               2 Dir(s)  68,658,302,976 bytes free

C:\LogFileExp>cd LogFileCountry

C:\LogFileExp\LogFileCountry>dir
 Volume in drive C is Windows
 Volume Serial Number is EA37-06AE

 Directory of C:\LogFileExp\LogFileCountry

26-04-2026  14:48    <DIR>          .
26-04-2026  14:49    <DIR>          ..
26-04-2026  14:48             1,668 LogFileCountryDriver.class
26-04-2026  14:44             1,112 LogFileCountryDriver.java
26-04-2026  14:48             1,755 LogFileMapper.class
26-04-2026  14:43               696 LogFileMapper.java
26-04-2026  14:48             1,619 LogFileReducer.class
26-04-2026  14:44               750 LogFileReducer.java
               6 File(s)          7,600 bytes
               2 Dir(s)  68,658,102,272 bytes free
```



Output:-

```
C:\LogFileExp>hdfs dfs -cat /LogFileExp/Output/part-00000
10.1.232.31     2
10.103.190.81   1
10.223.157.186  3
```



Weather.java

```
import java.io.IOException;
import java.util.ArrayList; import
java.util.Iterator; import java.util.List;
 import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
importorg.apache.hadoop.mapred.FileOutputFormat
; import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import rg.apache.hadoop.mapred.KeyValueTextInputFormat;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
import org.apache.hadoop.util.*;
/**
     * This is an Hadoop Map/Reduce application for Working on weather data It reads
     * the text input files, breaks each line into stations weather data and finds
     * average for temperature, dew point , wind speed. The output is a locally * sorted list of
       stations and its 12 attribute vector of average temp , dew , * wind speed of 4 sections for
       each month.
*/
public class Weather extends Configured implements Tool { final
    long DEFAULT_SPLIT_SIZE = 128 * 1024 * 1024;
    /**
     * Map Class for Job 1
     * For each line of input, emits key value pair with
     * station_yearmonth_sectionno as key and 3 attribute vector with
     * temperature , dew point , wind speed as value.Map method will strip the
     * day and hour from field and replace it with section no (
     * <b>station_yearmonth_sectionno</b>, <b><temperature,dew point , wind
     * speed></b>).
     */ public static class MapClass extends MapReduceBase implements
    Mapper<LongWritable, Text, Text, Text> {
        private Text word = new Text(); private
        Text values = new Text();
        public void map(LongWritable key, Text value,
                                OutputCollector<Text, Text> output,
                                Reporter reporter) throws IOException {
            String line = value.toString();
            StringTokenizer itr = new StringTokenizer(line);
            int counter = 0; Stringkey_out = null; String
            value_str = null; boolean skip = false;
```

```
loop:while (itr.hasMoreTokens() && counter<13) {
        String    str    =    itr.nextToken();    switch
        (counter) { case 0:
                key_out = str;
                if(str.contains("STN")){//Ignoring rows where stationid is all 9
                        skip     =     true;
                        break loop;
                }else{ break;
                }  case  2:  int  hour  =  Integer.valueOf(str.substring(str.lastIndexOf("_")+1,
        str.length())); str = str.substring(4,str.lastIndexOf("_")-2); if(hour>4 && hour<=10){
                        str = str.concat("_section1");
                }else if(hour>10 && hour<=16){ str =
                        str.concat("_section2");
                }else if(hour>16 && hour<=22){ str =
                        str.concat("_section3");
                } else{ str = str.concat("_section4");
                }

                key_out = key_out.concat("_").concat(str); break;
        case    3://Temperature    if(str.equals("9999.9")){//Ignoring    rows
                temperature is all 9
                        skip     =     true;
                        break loop;
                }else{
                        value_str = str.concat(" ");
                        break;
                }
        case  4://Dew  point  if(str.equals("9999.9")){//Ignoring  rows  where
                dewpoint all 9
                        skip     =     true;
                        break        loop;
                        }else{   value_str
                        =
                        value_str.concat
                        (str).concat("  ");
                        break;
                }
        case   12://Wind   speed   if(str.equals("999.9")){//Ignoring   rows
                wind speed is all 9 skip = true; break loop;
                }else{ value_str = value_str.concat(str).concat(" "); break;
                }
        default: break;
        }
        counter++;
} if(!skip){ word.set(key_out);
values.set(value_str);
output.collect(word, values);
}
    }
}
/**
 * Reducer Class for Job 1
```

```
 * A reducer class that just emits 3 attribute vector with average
 * temperature , dew point , wind speed for each of the section of the month for each input
 */ public static class Reduce extends MapReduceBase
implements Reducer<Text, Text, Text, Text> {
        private Text value_out_text = new Text(); public
        void reduce(Text key, Iterator<Text> values,
                        OutputCollector<Text, Text> output, Reporter reporter) throws IOException {
                double sum_temp = 0; double sum_dew = 0; double sum_wind= 0; int count = 0;

                while (values.hasNext()) {
                        String str = values.next().toString();
                        StringTokenizer itr = new StringTokenizer(str);
                        int     count_vector    =    0;    while
                        (itr.hasMoreTokens()) {
                                String    nextToken    =    itr.nextToken("    ");
                                if(count_vector==0){        sum_temp        +=
                                Double.valueOf(nextToken);
                                }
                                if(count_vector==1){        sum_dew        +=
                                        Double.valueOf(nextToken);
                                }
                                if(count_vector==2){        sum_wind        +=
                                        Double.valueOf(nextToken);
                                }
                                count_vector++;
                        }
                        count++;
                }
                double avg_tmp = sum_temp / count;
                double avg_dew = sum_dew / count;
                double avg_wind = sum_wind / count;
                System.out.println(key.toString()+" count is "+count+" sum of temp is "+sum_temp+" sum of
dew is "+sum_dew+" sum of wind is "+sum_wind+"\n");
                String        value_out        =        String.valueOf(avg_tmp).concat("
").concat(String.valueOf(avg_dew)).concat(" ").concat(String.valueOf(avg_wind));
                value_out_text.set(value_out);
                output.collect(key, value_out_text);
        }
}
static int printUsage() {
        System.out.println("weather [-m <maps>] [-r <reduces>] <job_1 input> <job_1 output> <job_2
output>");
        ToolRunner.printGenericCommandUsage(System.out); return
        -1;
}
/**
 * The main driver for weather map/reduce program.
 * Invoke this method to submit the map/reduce job.
 * @throws IOException When there is communication problems with the job tracker.
 */   public   int   run(String[]   args)   throws
Exception {
        Configuration config = getConf();
```

```
            // We need to lower input block size by factor of two
            JobConf conf = new JobConf(config, Weather.class);
            conf.setJobName("Weather Job1");

            // the keys are words (strings)
            conf.setOutputKeyClass(Text.class);
            // the values are counts (ints)
            conf.setOutputValueClass(Text.class)
            ;

            conf.setMapOutputKeyClass(Text.class);
            conf.setMapOutputValueClass(Text.class);
            conf.setMapperClass(MapClass.class);
            //conf.setCombinerClass(Combiner.class);
            conf.setReducerClass(Reduce.class);
            List<String> other_args = new ArrayList<String>();
            for(int  i=0;  i  <  args.length;  ++i)  {  try  {  if  ("-
            m".equals(args[i])) {
                                    conf.setNumMapTasks(Integer.parseInt(args[++i]));
                        }           else            if          ("-r".equals(args[i]))          {
                                    conf.setNumReduceTasks(Integer.parseInt(args[++i]));
                        } else {
                                    other_args.add(args[i]);
                        }
                  } catch (NumberFormatException except) {
                        System.out.println("ERROR: Integer expected instead of " + args[i]);
                        return printUsage();
                  } catch (ArrayIndexOutOfBoundsException except) {
                        System.out.println("ERROR: Required parameter missing from " +
                                    args[i-1]);
                        return printUsage();
                  }
            }
            // Make sure there are exactly 2 parameters left.
            FileInputFormat.setInputPaths(conf, other_args.get(0));
            FileOutputFormat.setOutputPath(conf,          new          Path(other_args.get(1)));
            JobClient.runJob(conf);
            return 0;
      }
      public  static  void  main(String[]  args)  throws  Exception  {  int  res  =
            ToolRunner.run(new   Configuration(),   new   Weather(),   args);
            System.exit(res);
      }
}
```

Input: sample_weather.txt (sample)

```
69019013910 20060201_0 51.75    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 000000
69019013910 20060201_1 54.74    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 00000
69019013910 20060201_2 50.59    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 00000
69019013910 20060201_3 51.67    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 000000
69019013910 20060201_4 65.67    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 000000
69019013910 20060201_5 55.37    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 000000
69019013910 20060201_6 49.26    33.024 1006.324    943.924 15.024    10.72422.0 28.9    0.00I999.9 000000
```

69019013910 20060201_7 55.44 33.024 1006.324 943.924 15.024 10.72422.0 28.9 0.00I999.9 000000
69019013910 20060201_8 64.05 33.024 1006.324 943.924 15.024 10.72422.0 28.9 0.00I999.9 000000

Output: part-00000.txt (on Hadoop)

690190_02_section1 53.87166666666666 25.899999999999995 7.774999999999998
690190_02_section2 54.76125000000001 25.900000000000006 7.774999999999999
690190_02_section3 53.25041666666667 25.899999999999995 7.774999999999996
690190_02_section4 52.44708333333333 25.900000000000006 7.774999999999999

Input:-

```
690190 13910 20060201_0 51.75 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_1 54.74 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_2 50.59 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_3 51.67 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_4 65.67 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_5 55.37 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_6 49.26 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_7 55.44 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
690190 13910 20060201_8 64.05 33.0 24 1006.3 24 943.9 24 15.0 24 10.7 24 22.0 28.9 0.001 999.9
000000
```

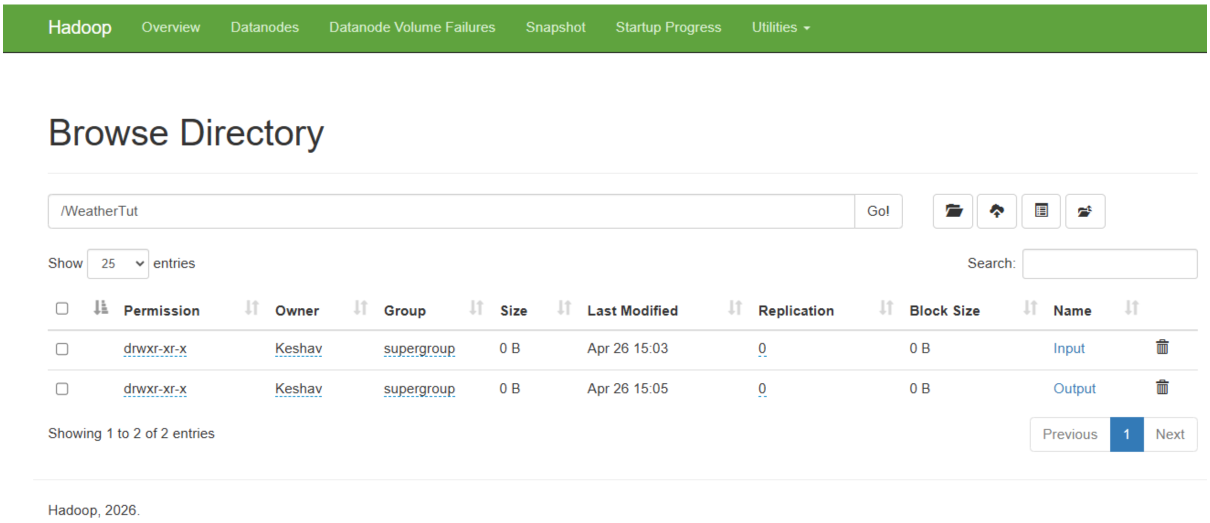
Algo:-

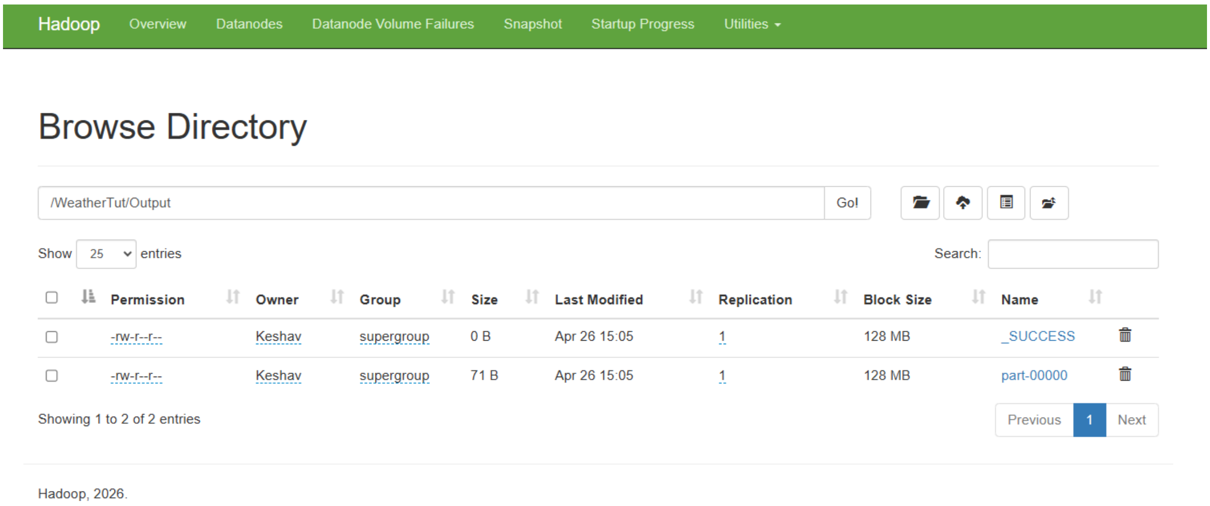
```
C:\WeatherExp>for /f "delims=" %i in ('hadoop classpath') do set HADOOP_CLASSPATH=%i

C:\WeatherExp>set HADOOP_CLASSPATH=C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common;C:\hadoop\share\ha
p\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\
hare\hadoop\common\lib\*;C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\had
n\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\etc\hadoop;C:\ha
are\hadoop\hdfs\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\y

C:\WeatherExp>
```

```
C:\WeatherExp>jar -cvf weathertut.jar Weather*.class
added manifest
adding: Weather$MapClass.class(in = 2940) (out= 1414)(deflated 51%)
adding: Weather$Reduce.class(in = 2994) (out= 1366)(deflated 54%)
adding: Weather.class(in = 3534) (out= 1806)(deflated 48%)
```

```
C:\WeatherExp>hadoop jar weathertut.jar Weather /WeatherTut/Input /WeatherTut/Output
2026-04-26 15:05:18,532 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-26 15:05:18,824 WARN mapreduce.JobResourceUploader: No job jar file set.  User classes may not
2026-04-26 15:05:18,924 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-26 15:05:19,229 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-26 15:05:19,583 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1341749057_000
2026-04-26 15:05:19,590 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-26 15:05:20,006 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-26 15:05:20,011 INFO mapreduce.Job: Running job: job_local1341749057_0001
2026-04-26 15:05:20,015 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-26 15:05:20,032 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOut
2026-04-26 15:05:20,103 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-26 15:05:20,106 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary fo
2026-04-26 15:05:20,181 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-26 15:05:20,184 INFO mapred.LocalJobRunner: Starting task: attempt_local1341749057_0001_m_00000
2026-04-26 15:05:20,211 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-26 15:05:20,211 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary fo
```

```
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=348
                Reduce input records=9
                Reduce output records=2
                Spilled Records=18
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=6
                Total committed heap usage (bytes)=371195904
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=925
        File Output Format Counters
                Bytes Written=71
```





Output:-

```
C:\WeatherExp>hdfs dfs -cat /WeatherTut/Output/part-00000
690190_02_section1      56.03 33.0 10.7
690190_02_section4      54.884 33.0 10.7
```

File information - part-00000

Download   Head the file (first 32K)   Tail the file (last 32K)

Block information -- Block 0

Block ID: 1073741834

Block Pool ID: BP-202349802-10.232.187.125-1777191315034

Generation Stamp: 1010

Size: 71

Availability:

- LAPTOP-GEJ3N0KM

File contents

```
690190_02_section1	56.03 33.0 10.7
690190_02_section4	54.884 33.0 10.7
```